METRIC

MIL-DTL-24582A(NAVY)
18 December 2018

SUPERSEDING
DOD-A-24582(NAVY)
25 October 1978

DETAIL SPECIFICATION

ANCHOR, TWO-FLUKE, BALANCED FLUKE

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high holding power, nonfouling, two-fluke, balanced fluke type anchors for use on ships and boats.

1.2 Classification. The anchors are of the following nominal masses, as specified (see 6.2):

Nominal masses kg (lbs)		
325 (660)	4,000 (8,000)	8,500 (17,200)
450 (925)	4,500 (8,930)	9,500 (19,200)
1,000 (1,985)	5,000 (10,000)	11,000 (21,800)
1,400 (2,800)	6,000 (11,600)	12,000 (24,500)
2,250 (4,600)	6,500 (13,200)	13,000 (27,500)
3,000 (6,000)	7,500 (15,200)	15,000 (30,400)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 2040

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:07Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application

MIL-DTL-22200/1 - Electrodes, Welding, Mineral Covered, Iron-Powder, Low-Hydrogen Medium and High Tensile Steel, As-Welded or Stress-Relieved Weld Application

MIL-PRF-23236 - Coating Systems for Ship Structures

MIL-PRF-24635 - Coating Systems, Weather-Resistant, Exterior Use

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-5000920 - Two-Fluke, Balanced Fluke Anchor Assembly and Details

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AQ-GIB-010/248 - Requirements for Welding and Brazing Procedure and Performance Qualification

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ [by searching for the document number without the suffix]. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:07Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A27/A27M	-	Standard Specification for Steel Castings, Carbon, for General Application
ASTM A36/A36M	-	Standard Specification for Carbon Structural Steel
ASTM A108	-	Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A370	-	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM A668/A668M	-	Standard Specification for Steel Forgings, Carbon and Alloy, for General Industrial Use
ASTM B103/B103M	-	Standard Specification for Phosphor Bronze Plate, Sheet, Strip, and Rolled Bar
ASTM E112	-	Standard Test Methods for Determining Average Grain Size

(Copies of these documents are available online at www.astm.org.)

NACE INTERNATIONAL

NACE No. 2/SSPC-SP 10	-	Near-White Metal Blast Cleaning
NACE WJ-2/SSPC-SP WJ-2	-	Waterjet Cleaning of Metals – Very Thorough Cleaning (WJ-2)

(Copies of these documents are available online at www.nace.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Shank, crown, and flukes of all anchor masses and ball guide for anchor masses of 4,000 kilograms (kg) (8,000 pounds [lbs]) and over. Shank, crown, and flukes of all anchor masses and ball guide for anchor masses of 4,000 kg (8,000 lbs) and over shall be steel castings in accordance with ASTM A27/A27M. Castings shall be heat-treated by normalizing and tempering to obtain the mechanical properties listed in table I. The detailed procedures to be used shall be determined by the manufacturer. The microstructure specimen shall have a minimum grain size of 6 in accordance with ASTM E112. Material having a grain size number less than 6 (larger grain size) shall be subject to rejection.

TABLE I. Mechanical properties.

Mechanical properties	Minimum
Tensile strength	4,921 kg/cm^2 (70,000 lb/in^2)
Yield point	3,164 kg/cm^2 (45,000 lb/in^2)

3.1.1.1 Heat treatment. In addition to the requirements of ASTM A27/A27M, castings may be reheat-treated by austenitizing, quenching, and tempering, or normalizing and tempering as necessary to attain the required physical properties.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:07Z
Check the source to verify that this is the current version before use.

3.1.1.2 Shank, crown, fluke, and ball guide test block. A test block shall be integrally cast with the shank, crown, flukes, and ball guide (unless damage to parts could result, in which case a separate cast test block from the heat poured with the casting shall be required) to comply with the chemical, mechanical, and microstructure test requirements. Adequate measures shall be taken to ensure test block soundness. Chills or chill molds shall not be used for the test block. The size of the test block shall be in proportion to the thickness (T), where T is the diameter of the largest sphere (±12.7 millimeters [mm] [±0.50 inch]) that can be inscribed in any cross section of the casting. Test blocks shall remain attached to the casting they represent until submitted for inspection. Test blocks shall be heat-treated with the casting. The minimum dimensions of the test block shall be T by T by 2T. Minimum T shall be 101.6 mm (4 inches) for all size anchors. Dimensions in excess of these shall be permitted providing sound test blocks are produced. Mechanical testing (see 4.5.2) shall be taken at least 25.4 mm (1 inch) below the test block surface.

3.1.2 Shackles, shackle pins, guide lug, lock pin, and clevis for all anchor masses and ball guide for anchor masses of 3,000 kg (6,000 lbs) and under. Shackles, shackle pins, guide lug, lock pin, and clevis for all anchor masses and ball guide for anchor masses of 3,000 kg (6,000 lbs) and under shall be forged steel in accordance with ASTM A668/A668M, class D, normalized and tempered. For ball guides and for anchors above 3,000 kg (6,000 lbs), see 3.1.1.

3.1.2.1 Test blocks. Test blocks for forgings shall be made from prolongs in accordance with ASTM A370. Specimens shall be taken at ¼T of prolong with the prolong having minimum dimensions of T by T by W, where T and W are the thickness and width, respectively, of the largest cross section.

3.1.3 Anchor shaft. The anchor shaft shall be made from steel bar conforming to AISI 1141 hot-rolled steel of 4,921 kilograms per centimeter squared (kg/cm^2) (70,000 pounds per square inch [lb/in^2]) yield strength.

3.1.4 Clevis pin. The clevis pin shall be made from steel bar in accordance with ASTM A108 with a maximum carbon content of 0.25.

3.1.5 Eye swivel, nut swivel, and swivel pin. The eye swivel, nut swivel, and swivel pin shall be made from AISI 4340 alloy steel , in accordance with ASTM A668/A668M, and shall be heat-treated after machining to obtain 302/321 Brinell hardness number (BHN). Threading of the eye swivel and nut swivel shall be in accordance with FED-STD-H28.

3.1.6 Washer. The washer shall be made of phosphor bronze, composition A hard, in accordance with ASTM B103/B103M.

3.1.7 Base plate, stowage pad, closing plate, and filler piece. The base plate, stowage pad, closing plate, and filler piece shall be steel in accordance with ASTM A36/A36M.

3.2 Description. The anchor assembly shall consist of a crown with a pair of sharp, double-swinging flukes, shank, shackle, the necessary connecting shaft, pins, and, when specified (see 6.2), a ball guide assembly. The anchor crown shall be hollow, and weights may be welded in its interior to achieve the specified nominal anchor mass. The crown shall have provisions for draining.

3.2.1 Anchor assembly. Each anchor shall be furnished complete (see 6.2), ready for use, with all parts properly fitted and assembled. The rounded shackle (see 3.3, piece [PC] 20) shall be used when the anchor is stowed above the waterline, while the square shackle (see 3.3, PC3) and ball guide assembly (see 3.2.2) shall be used for anchor stowage below the waterline.

3.2.2 Ball guide assembly. The ball guide assembly shall consist of the ball guide, guide lugs, clevis, eye and nut swivel, along with the necessary connecting and locking pins (see 3.3).

3.3 Design and dimensions. The anchor and ball guide assemblies, including shackle and pin, if applicable, shall comply with design and dimensions shown on Drawing 803-5000920.

3.4 Mass tolerances. The mass tolerances on the anchor assembly shall not vary more than +4 percent to -2 percent of the nominal mass specified (see 3.3). These tolerances are not applicable to the ball guide assembly.

3.5 Magnetic particle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:07Z
Check the source to verify that this is the current version before use.

3.5.1 Castings. For castings, all linear indications in accordance with MIL-STD-2035 that are over 6.35 mm (0.25 inch) in length shall be ground to sound metal and well faired with the surrounding surface. Weld repair shall not be required if remaining thickness of the original metal equals or exceeds dimensions shown on the drawing (see 3.3). Depth of the grinding shall be not greater than 38.1 mm (1.5 inches), except for cracks which shall be removed fully. Casting may be built up to required dimensions by weld repair.

3.5.2 Forgings. Forgings shall comply with class 3 of MIL-STD-2035. Indications in excess of class 3 shall be ground to sound metal and well faired with surrounding surfaces.

3.5.3 Qualification of welders and welding procedure. Qualification of welders and welding procedure shall be accomplished in accordance with T9074-AQ-GIB-010/248. All final welds shall meet the requirements of MIL-STD-2035, class 2.

3.5.4 Repair welding. Unless otherwise specified (see 6.2), defects in castings and forgings may be removed and repaired by welding. Weld repairs shall use MIL-7018-M electrodes in accordance with MIL-DTL-22200/l. Prior to repair welding, the excavated area shall be fully visible to the welder and allow access of the electrode to all weld surfaces.

3.5.4.1 Minor weld repairs. Minor weld repairs may be made to correct surface defects provided that:

a. they do not exceed 25 mm (0.98 inch) in depth or 20 percent of material thickness, whichever is less

b. a weld buildup for correction of dimensional or machining errors does not exceed the following limitations:

(1) 4.75 mm (0.187 inch) maximum buildup for design thickness of 25.4 mm (1 inch) and under

(2) 20 percent maximum buildup for design thickness over 25.4 mm (1 inch), but not to exceed 9.52 mm (0.375 inch) unless approved

(3) contouring (such as grinding or machining) for discontinuities shall blend into adjacent material and shall not reduce the thickness of the material below minimum design requirements

3.5.4.2 Nominal weld repairs. Repair welds in excess of those specified as minor repairs shall be made only with the approval of the Government representative.

3.6 Proof load. Anchors shall comply with the proof load test specified in 4.5.3. There shall be no permanent deformation of any part of the anchor upon completion of this test.

3.7 Cleaning. The castings and forgings shall be cleaned for proper adhesion of finish when offered for inspection.

3.8 Identification markings. Shanks shall bear the legend specified on the applicable drawing (see 3.3).

3.9 Finish. Surfaces shall be cleaned in accordance with SSPC-SP 1 to remove oil, grease, and residual contaminants. Anchors surfaces shall be prepared with near white metal blast in accordance with NACE No. 2/SSPC-SP 10. Waterjetting to NACE WJ-2/SSPC-SP WJ-2 may be used in place of NACE No. 2/SSPC-SP 10. Anchors shall have a primer of one coat in accordance with MIL-PRF-23236, type V or VI, class 5 or 7, 4 to 8 mils; or one coat in accordance with MIL-PRF-23236, type VII, class 5 or 7, 4 to 8 mils. Anchors shall have a top coat of one coat in accordance with MIL-PRF-23236, type V or VI, class 5 or 7, 4 to 8 mils; or one coat in accordance with MIL-PRF-23236, type VII, class 5 or 7, 10 to12 mils. Anchors shall have one coat of haze gray in accordance with MIL-PRF-24635; type III, V, or VI; class 1; grade B or C; 2 to 3 mils.

3.10 Preservation. After welding of the base plate to the bottom of the anchor, the interior section of the anchor shall be filled with a rust preventative compound in accordance with MIL-PRF-16173, grade 1 and then drained of the excess compound.

3.11 Workmanship. The workmanship shall be free from imperfections that may impair appearance or serviceability.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:07Z
Check the source to verify that this is the current version before use.

4.2 Sample lots.

4.2.1 Castings. A lot shall consist of castings made from the same heat. Where material cannot be identified by heat, it shall be divided into lots of not more than 500 kg (1,000 lbs) each, except where a single casting has a mass of more than 500 kg.

4.2.2 Forgings. A lot shall consist of material manufactured from the same heat.

4.2.3 Chemical analysis of forgings. A lot shall consist of all forgings from the same heat.

4.2.4 Chemical analysis of castings. A lot shall consist of castings of one class made from the same heat, heat-treated in the same furnace charge, and offered for inspection at one time.

4.2.5 Assembled anchor. A lot shall consist of assembled anchors of the same size offered for inspection at the same time.

4.3 Sampling for lot acceptance.

4.3.1 Sampling for visual and dimensional examination. Sampling for visual and dimensional examination and lot acceptance criteria shall be in accordance with table II.

TABLE II. Sampling for visual and dimensional examination.

Number of anchors in inspection lot	Number of anchors in sample	Acceptance number (defectives)
25 and under	10	0
26 to 40	15	1
41 to 110	25	1
111 to 300	35	2
301 to 500	50	3
501 to 800	75	4
801 and over	110	6

4.3.2 Sampling for nominal mass. Sampling for nominal mass examination and lot acceptance criteria shall be in accordance with table III.

TABLE III. Sampling for nominal mass tests, assembled anchors tolerance.

Number of anchors in inspection lot	Number of anchors in sample	Acceptance number (defectives)
40 and under	5	0
41 to 110	10	0
111 to 300	15	1
301 to 500	25	1
501 to 800	35	2
801 and over	50	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:07Z
Check the source to verify that this is the current version before use.

4.3.3 Sampling for magnetic particle test. Sampling for the magnetic particle test shall be 100 percent for all parts except that pins, base plate, closing plate, and the phosphor bronze washer shall not be subjected to this test.

4.3.4 Sampling for chemical analysis of castings. Where material cannot be identified by heat, it shall be divided into lots of not more than 500 kg (1,000 lbs) each, except where a single casting has a mass of more than 500 kg. One sample shall be obtained from each lot for chemical analysis and these samples shall be analyzed individually.

4.3.5 Sampling for chemical analysis of forgings. Samples for chemical analysis of forgings shall be taken from the same lot that consists of material manufactured from the same heat.

4.3.6 Sampling for tension test and microstructure tests on castings. For castings estimated to have a mass of 250 kg (500 lbs) and over, one sample for the tension test shall be taken to represent each casting. For castings estimated to have a mass less than 250 kg (500 lbs), two samples for the tension test shall be taken from each lot. Microstructure examination for castings shall be performed on a specimen taken from the test block adjacent to the tensile specimen and away from the surface of the test block.

4.3.7 Sampling for tension tests on forgings. For all forged parts, except shackles and pins from anchors having a mass of 500 kg (1,000 lbs) and less, two samples for the tension test shall be taken from each lot.

4.4 Examinations.

4.4.1 Visual and dimensional examination. Anchors selected in accordance with table II shall be visually and dimensionally examined, both prior to and after painting, for defects in manufacturing and workmanship to verify compliance with this specification. Any anchor containing one or more visual or dimensional defects shall be rejected and if the number of defective anchors in any sample exceeds the acceptable number for that sample, the entire lot represented by the sample shall be rejected. Rejected lots may be offered again for inspection, provided all nonconforming anchors have been removed. Samples shall again be selected from such rejected lots and re-examined to verify compliance with this specification.

4.4.2 Shank. Radii on the corners of the shanks shall be examined to ensure a smooth curve free from irregularities. The stop on the shank shall make full surface contact with the boss in the crown when the flukes are in the operating position.

4.4.3 Determination of mass. Each anchor of the sample shall have its mass determined to verify compliance with 3.3. Anchors exceeding the tolerance limit specified on the drawing (see 3.3) shall be rejected. If the number of defective anchors in any sample (see table III) exceeds the acceptable number for that sample, the entire lot represented by that sample shall be rejected. Rejected lots may be resubmitted for inspection only after all nonconforming anchors have been removed. Samples shall again be selected from such rejected lots and re-examined to verify compliance with this requirement.

4.5 Test procedures.

4.5.1 Chemical tests. Samples selected in accordance with 4.3.4 and 4.3.5 shall be analyzed in accordance with ASTM A27/A27M or ASTM A668/A668M, whichever is applicable, to ensure conformance with chemical composition requirements. The methods shall be correlated with the National Institute of Standards and Technology standard reference material to ensure validity of the method that is used as a control in chemical methods of analysis or for calibration in instrumental methods of analysis. The specification covering the method of analysis used shall be noted on certification documents.

4.5.2 Mechanical property tests. Tension tests shall be conducted in accordance with ASTM A370.

4.5.3 Proof load test.

4.5.3.1 General. Each anchor shall be assembled with shackle, pins, and ball guide assembly (if applicable), ready for service and shall be subjected, in the presence of the Government representative, to the proof load test in both positions as specified on the applicable drawing (see 3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:07Z
Check the source to verify that this is the current version before use.

4.5.3.2 Gaging. The anchors shall be gaged before and after they have been tested in each operative position to determine whether any permanent deformation has been caused by applied stress.

4.5.4 Microstructure test. Microstructure examination shall be in accordance with ASTM E112.

4.5.5 Magnetic particle test. Magnetic particle examination shall be performed following completion of proof load testing and shall be in accordance with T9074-AS-GIB-010/271. The anchor proper, the ball guide, and all components of the anchor and ball guide assemblies shall be subjected to magnetic particle examination. Examinations shall be performed after proof loading from both sides and prior to any coating or painting. Special attention shall be given to the following:

a. fluke surfaces from tip up to and including radii at junction with shaft
b. junction where the crown-fluke piece is joined to the shank by the shaft
c. crown bearing area that acts as a stop for the shank
d. shackle pin area and shank in the vicinity of the shackle pin
e. upper and lower attachment points of the ball guide and areas where marked changes in cross section occur
f. the shackle at bend and pin hole
g. all welds in the anchor and ball guide assemblies, except in the swivel eye nut and pin areas

Repair welds shall be subjected to the same inspection standards required of the casting and the area shall be suitably marked to facilitate inspection. Anchors that are repair welded shall be proof load tested afterwards.

Any part in the sample containing one or more defects shall be rejected. Rejected lots may be resubmitted for inspection only after every anchor part has been examined for the type of defect found and all defective parts have been removed from the lot.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Anchors covered by this specification are for use on U.S. naval ships. Ball guide assemblies must be used for only those anchors stowed below the waterline. Two-fluke, balanced fluke anchors made to this specification are a design with known performance characteristics and geometric properties that are compatible with U.S. Navy ships. Existing stock of anchors manufactured to DOD-A-24582 dated 25 October 1978 is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Nominal mass of anchor required (see 1.2).
c. Ordering options:
   (1) Complete anchor with ball guide assembly (see 3.2).
   (2) Complete anchor assembly (without ball guide assembly (see 3.2.1).
   (3) Ball guide assembly – order by nominal mass of anchor (see 3.2.2).
d. Repair welding if other than specified (see 3.5.4).
e. Packaging requirements (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:07Z
Check the source to verify that this is the current version before use.

6.3 Applicability. Anchors covered by this specification are commonly referred to as balanced fluke anchors.

6.4 Subcontracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for use or incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Anchor shaft
Ball guide
Base plate
Clevis
Clevis pin
Closing plate
Crown
Eye swivel
Filler piece
Guide lug
High holding power
Lock pin
Nonfouling
Nut swivel
Shackle
Shackle pin
Shank
Stowage pad
Swivel pin
Test block
Washer

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 2040-2019-002)

Review activity:
Navy – YD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:07Z
Check the source to verify that this is the current version before use.